Speaker Information

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Talk Information

Behavioral persistence maintained by *C. elegans* GPCR kinase GRK-1 in a dynamic thermal environment

Animals often encounter fluctuating sensory inputs as they navigate complex environments. How the nervous system stabilizes behavior over time despite such sensory variability remains poorly understood. In this talk, I will introduce a neural mechanism that sustains a stable forward locomotor state specifically at temperatures associated with prior food experience in the nematode *Caenorhabditis elegans*. My previous work with Professor Ikue Mori and colleagues showed that *C. elegans* maintains prolonged temperature tracking at the cultivation temperature by simultaneously inhibiting two head interneurons, RIM and AVA. Furthermore, we found that a worm homolog of human GPCR kinase GRK5 functions cell-autonomously in both RIM and AVA interneurons to maintain sustained isothermal tracking. Our findings reveal how a GPCR kinase acts within defined neurons to stabilize a behavioral state over time in animals freely moving in a sensory environment. Given the conserved and broad expression of GRK-1/GRK5 in the nervous systems of worms and mammals, our work provides insights into understanding persistent behavioral states in more complex nervous systems and environments.

Research Overview (for collaboration discussion)

[Research Interests] Genetic and neural basis of adaptive behaviors

[Current Research Projects] Olfactory learning in the fruit fly *D. melanogaster*

[Techniques] Behavioral genetics, animal tracking and neuronal imaging

[Aspects or techniques of interest for future collaboration]

Comparative studies of adaptive behaviors from model and non-model animals